A BLOW-UP SQUARE THAT IS A PUSHOUT WITH AN INJECTIVE LEG PRODUCES A DISTINGUISHED TRIANGLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the reduction to injective sheaves is stated, and exactness at each of the three positions of the displayed sequence is proved, including the amalgamation step in the middle.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The canonical map of a blow-up square. Same chapter, Lemma lemma-c-square. Supplies the construction whose output is shown to be a distinguished triangle.
- K-injective representatives in the unbounded derived category. Same chapter, Section section-unbounded. Allows the given complex to be replaced by a K-injective complex of injective sheaves.
- Free abelian sheaves represented by sheafified Yoneda functors. Modules on Sites, Section sites-modules-section-free-abelian-sheaf. Supplies the description of the values of an injective sheaf as homomorphisms from the sheafified representable functor.

Source reference. Original lemma 0F16: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6001–6001. Navigation pointer invoked at proof line 6080; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 07A5: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3921–3921. Navigation pointer invoked at proof

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line 6093; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 03CQ: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6112–6112. Navigation pointer invoked at proof line 6112; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Consider a commutative diagram

$$\begin{array}{ccc} E & \longrightarrow & Y \\ \downarrow & & \downarrow \\ Z & \longrightarrow & X \end{array}$$

in the category $\mathcal{C}$. In this situation, given an object K of $D(\mathcal{O})$ we get what looks like the beginning of a distinguished triangle

$$R\Gamma(X, K) \to R\Gamma(Z, K) \oplus R\Gamma(Y, K) \to R\Gamma(E, K)$$

In the following lemma we make this more precise.

Lemma 0.1. *In the situation above assume*

(1) $h_X^\# = h_Y^\# \amalg_{h_E^\#} h_Z^\#$, *and*

(2) $h_E^\# \to h_Y^\#$ *is injective.*

Then the construction of Lemma 0F16 produces a distinguished triangle

$$R\Gamma(X, K) \to R\Gamma(Z, K) \oplus R\Gamma(Y, K) \to R\Gamma(E, K) \to R\Gamma(X, K)[1]$$

functorial for K in $D(\mathcal{C})$.

Proof. We can represent K by a K-injective complex whose terms are injective abelian sheaves, see Section 07A5. Thus it suffices to show: if $\mathcal{I}$ is an injective abelian sheaf, then

$$0 \to \mathcal{I}(X) \to \mathcal{I}(Z) \oplus \mathcal{I}(Y) \to \mathcal{I}(E) \to 0$$

is a short exact sequence. The first arrow is injective because by condition (1) the map $h_Y \amalg h_Z \to h_X$ becomes surjective after sheafification, which means that $\{Y \to X, Z \to X\}$ can be refined by a covering of X. The last arrow is surjective because $\mathcal{I}(Y) \to \mathcal{I}(E)$ is surjective. Namely, we have $\mathcal{I}(E) = \mathrm{Hom}(\mathbf{Z}_E^\#, \mathcal{I})$, $\mathcal{I}(Y) = \mathrm{Hom}(\mathbf{Z}_Y^\#, \mathcal{I})$, the map $\mathbf{Z}_E^\# \to \mathbf{Z}_Y^\#$ is injective by (2), and $\mathcal{I}$ is an injective abelian sheaf. Please compare with Modules on Sites, Section 03CQ. Finally, suppose we have $s \in \mathcal{I}(Y)$ and $t \in \mathcal{F}(Z)$ mapping to the same element of $\mathcal{I}(E)$. Then s and t define a map

$$s \amalg t : h_Y^\# \amalg h_Z^\# \longrightarrow \mathcal{I}$$

which by assumption factors through $h_Y^\# \amalg_{h_E^\#} h_Z^\#$. Thus by assumption (1) we obtain a unique map $h_X^\# \to \mathcal{I}$ which corresponds to an element of $\mathcal{I}(X)$ restricting to s on Y and t on Z. $\square$